Body Mass Index (BMI) estimation from a single frontal face image using a Bayesian-ViT architecture

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Abstract

Body Mass Index (BMI) is an attribute that can be used for a number of applications ranging from surveillance, identification, image retrieval systems and healthcare, amongst others. Healthcare applications of BMI have gained special importance for detecting people at high risk of suffering malnutrition, obesity, diabetes or cardiovascular problems. This work presents a Bayesian-ViT architecture to estimate BMI from a single face image. The model was trained using a benchmark dataset from literature (VisualBMI) and compared with state of the art algorithms. Results show that the proposed method is comparable to reported methods achieving an overall MAE of 4.434. The Bayesian fashion of the model allows uncertainty to be accounted for. Therefore, results can be selected according to the predicted variance. Using uncertainty, the model achieve MAE values between 2.560 and 3.668 when selecting the best 10% and 75% of the data respectively.

Keywords: Bayesian-ViT, Body Mass Index (BMI), Uncertainty estimation

1 Introduction

Body Mass Index (BMI) is a metric that uses a person's height and weight to estimate their body fat. It is calculated as weight in kilograms divided by the square of height in meters ($kg \div m^2$). The resulting value is usually classified in several categories from underweight to obese. The world health organization proposes the following classes: Normal weight (BMI $\leq$ 25), Overweight (25 $<$ BMI $\leq$ 30) and Obese I (30 $<$ BMI $\leq$ 35) and Obese II (BMI $>$ 35). This is a basic indicator that is broadly used to categorize people's health. It is specially useful to detect potential health risks related to weight such as malnutrition, obesity, cardio-vascular diseases or diabetes.

The fast development of computer vision techniques and deep learning in recent years have allowed the use of single face images to estimate BMI. This is mainly due to the correlation between facial geometry and Body Mass. Such techniques provide automatic and non-invasive tools that become very useful for detecting people at high risk of health disorders due to weight such as malnutrition or obesity. Obesity, in particular, has tripled worldwide since 1975, with 13% of adults currently obese and 39% overweight globally. This is directly linked to serious conditions including type 2 diabetes, cardiovascular disease, several cancers, sleep apnea, osteoarthritis, depression and mental health disorders amongst others.

This work, combines state of the art Vision Transformer architecture with Bayesian inference, to estimate BMI from single face images. It includes uncertainty measurements to analyse the results of the Bayesian framework and to improve estimation results. The remaining of the paper is organised as follows: Section 2 presents the most relevant methods for estimating BMI from images and the benchmark dataset used the most in the literature. Section 3 describe the Bayesian ViT architecture implemented while Section 4 reports the results obtained and compares them with the state of the art. Finally, Section 5, reports the conclusion and future work.

2 State of the Art

Computer vision techniques, such as Deep learning, have allowed BMI to be estimated directly from images promoting non invasive systems than can be used remotely. Several methods have been reported, mainly differing on the input data, the architecture and the dataset used. The most commonly used input data are 2D and 3D full-body images [Manichand et al., 2025, Pentakota et al., 2024], multi-view images/videos [Kim et al., 2023, Xiang et al., 2025] and single frontal facial image. The latter being the focus of this paper.

Facial images have been studied in the literature as a key source of information including age, gender, ethnicity, identity, blood pressure, stress level, alcohol consumption, genetic conditions, and weight amongst others [Agbo-Ajala and Viriri, 2021, Dantcheva et al., 2018, Pal et al., 2025]. Therefore, it is not surprising that BMI can also be estimated from face images since it has demonstrated a correlation with facial geometry. Kocabet *et al.* [Kocabey et al., 2017] were some of the first to introduce BMI estimation from a single face image. They proposed a regression method based on the 50-layers ResNet-architecture and suggested that facial images contain discriminatory information pertaining to height, weight and BMI, comparable to that of full-body images and videos.

Since then, several works have been reported in this field (See Table 1). The most well known benchmark datasets in the area are MORPH-II [Ricanek and Tesafaye, 2006], VisualBMI [Kocabey et al., 2017], VIP-Attribute and FIW-BMI [Jiang et al., 2019] and Illinois DOC dataset [Fisher, 2019]. Table 1 shows the best results obtained from each data set. The differences of MAE between datasets are mainly due to bias in race, gender or data imbalance. Siddiqui et al.[Siddiqui et al., 2022] have shown that black males obtained the lowest MAE across different CNN-based architectures. This results shown in table Table 1 where the database with the best MAE results is Morph-II which corresponds to images from the African-American community (with a majority of male subjects). The same study has shown that white females, on the other hand, obtained the worst MAE amongst all the gender-race groups .

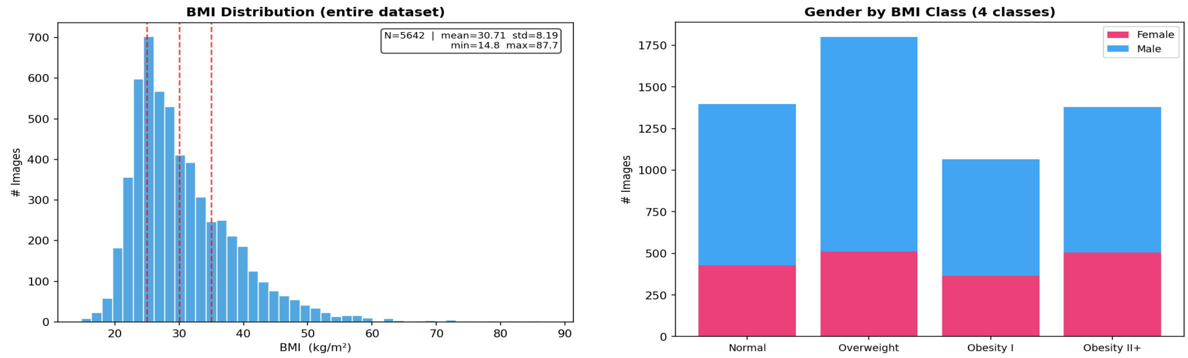
Another factor that influence the global BMI estimation results reported in the literature is the imbalance of data regardless of the BMI category (Underweight, Normal, overweight and obese). Most databases do not include the Underweight category and datasets like VIP-attributes tends to over represent only a certain type of population. Since such databases were created from celebrity images which in general tend to have lower BMI than the general population.

Paper	MAE	Dataset
[Kocabey et al., 2017]	5.16	VisualBMI
[Haritosh et al., 2019]	3.80	
[Yousaf et al., 2021]	4.99	
[Siddiqui et al., 2022]	5.02	
[Aarotale et al., 2023]	5.13	
[Jiang et al., 2020]	2.61	Morph-II
[Yousaf et al., 2021]	1.72	VIP attributes
[Jiang et al., 2019]	3.70	FIW-BMI
[Sidhpura et al., 2022]	3.22	Illinois DOC

Table 1: Summary of the state of the art techniques for BMI estimation from single face image

In this work, a Bayesian architecture based on Transformer models is proposed for the estimation of BMI from single facial images. The Transformer architecture was first introduced by Dosovitskiy *et al.* [Dosovitskiy et al., 2020] and since then, it has been applied to a wide range of problems in computer vision.

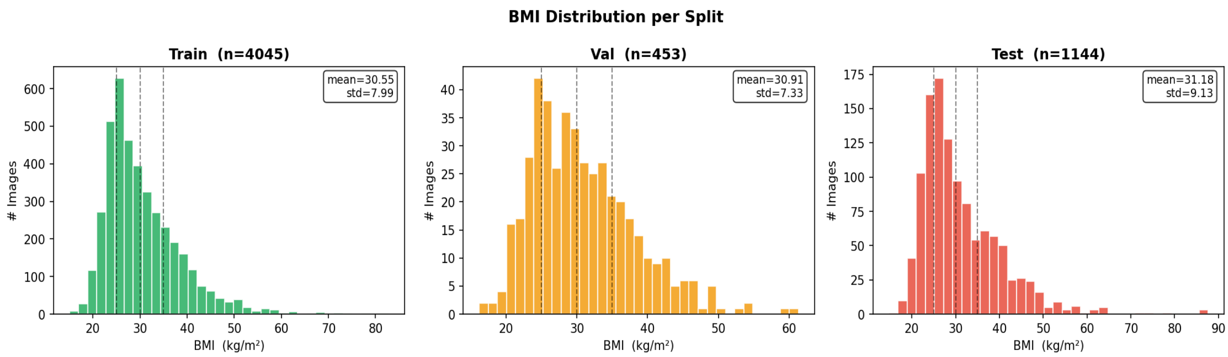
Bayesian Neural Networks [Abdar et al., 2021], on the other hand, is a popular approach to model uncertainty. Given a set of inputs $\{x_1, x_2, x_3, ...x_n\}$ with their outputs $\{y_1, y_2, y_3, ...y_n\}$ it is possible to model a function f such that $y = f(\mathbf{x})$. A prior distribution over the space of functions $p(f)$ can be used as prior belief. The likelihood can then be defined as $p(y|f, \mathbf{x})$ and using the Bayes rule, the posterior found $p(f|\mathbf{x}, y)$ given the input dataset $\{\mathbf{x}, y\}$. The output is then predicted for a new input x^* by integrating over all possible functions f. Since the posterior is, in general, intractable, variational inference is used to approximate it with a tractable distribution $q(\mathbf{w})$ by minimizing the Kullback-Leibler (KL) divergence. Bayesian modelling has proven to be an useful tool for estimating uncertainty in a wide range of fields ranging from medicine [Ngartera et al., 2024] to agronomy [Arellano et al., 2025]



(a) Dataset Histogram

(b) Samples per classes

Figure 1: Distribution of the data sets used. In (a) the histogram of the BMI is reported while in (b) The images for each class according to the World Health Organization (WHO) and displayed by gender.



(a) BMI in Train Dataset

(b) BMI in Validation Dataset

(c) BMI in Test Dataset

Figure 2: BMI Distribution of the datasets used for training (a), validation (b) and testing (c) . The number of images, the mean and the standard deviation of the BMI distribution is displayed in each figure.

3 Proposed Method

This paper proposes a Bayesian-ViT architecture to estimate BMI from single frontal images. The model has been trained and tested using the benchmark VisualBMI dataset [Jiang et al., 2019]. The dataset and the implementation details of the model are described as follows.

3.1 Dataset

The original VisualBMI data set was collected from the Reddit social media platform. Users reported images before and after weight changes along with their height and weight both before and after. The BMI was then computed for each subject from the reported height and weight values. This data set was originally composed of full body images. For this reason, a Multi-task Cascaded Convolutional Network (MTCNN) was used to detect and crop-out faces.

A total of 5,897 annotated images were available in the public Kaggle release of VisualBMI. After cropping faces and eliminating duplicates, only 5,642 images were considered. Figure 1(a) shows the BMI distribution of the dataset while Figure 1(b) shows the BMI distribution per class and gender. Where the classes are defined according to the World Health organization as: Normal, Overweight, Obesity I and Obesity II+.

The dataset was split in three sets of images: training (4,045), validation (453) and test (1,144). Two considerations were taken into account when splitting the dataset: (1) To preserve the BMI distribution along datasets and (2) to ensure that the datasets are person-level disjoint. Figure 2 shows the distribution of the

Method	Normal	Overweight	Obesity I	Obesity II	Male	Female	Overall
ViT	3.592	2.516	3.275	7.818	4.037	5.098	4.375
Bayesian-ViT	3.361	2.497	3.400	8.196	4.073	5.209	4.434

Table 2: MAE value obtained for each method (ViT and Bayesian-ViT) computed by class (Normal, Overweight, Obesity I and Obesity II) and gender (Male and Female). The final column corresponds to the MAE values computed over all images in the testing dataset for each method.

resulting datasets while Figure 3 shows an example of images taken from the dataset.

3.2 Bayesian-ViT Model

The proposed Bayesian-ViT Model was built using a ViT-B/16 Vision Transformer as backbone [Dosovitskiy et al., 2020]. This model uses a 16×16 patch size and it was first pre-trained using the ImageNet-21k dataset ($14M$ images, $21,000$ classes). An additional fine-tuning process with the ImageNet-1k dataset ($1.28M$ images, $1,000$ classes) was also performed using the method described by [Steiner et al., 2021]. Two variational layers were added (at the top) to make the regression model Bayesian. These two layers had a double number of parameters with respect to a non variational layer since the parameters to be estimated were distributions. Therefore, each parameters was defined by its mean and standard deviation. The estimation output also became a predictive distribution $p(\hat{y} \mid \mathbf{x}) = \mathcal{N}(\mu^*, \sigma^{2*})$ of the BMI with mean μ^* and standard deviation σ^*. The variance (σ^{2*}), therefore, acted as a metric of uncertainty, where small variance will indicate a more trusted result. In addition to the variance, the entropy was computed as follows:

Figure 3: Random samples of facial images from the VisualBMI dataset.

$$\sigma^{2*} = \frac{1}{T}\sum_{t=1}^{T}\left(\hat{y}^{(t)} - \mu^*\right)^2; \qquad H[\,p(\hat{y} \mid \mathbf{x})\,] = \frac{1}{2}\ln\left(2\pi\, e\, \sigma^{2*}\right) \tag{1}$$

Where T corresponded to the number of stochastic forward passes performed in the model (set to 30 in all the experiments). Lower entropy indicated more confident predictions. Those metrics were used to analyse the uncertainty of the predictions. The following section (Section 4) presents the experiments performed and the analysis of the performance of the model where the uncertainty was used to select the best BMI estimation results.

4 Experiments and Results

Experiments performed and the results of BMI estimation using ViT architecture with and without the Bayesian components are presented in section 4.1. Section 4.2, on the other hand, reports the performance of the model when uncertainty is used to select the best results. A comparison of the results with state of the art techniques is reported in Section 4.3.

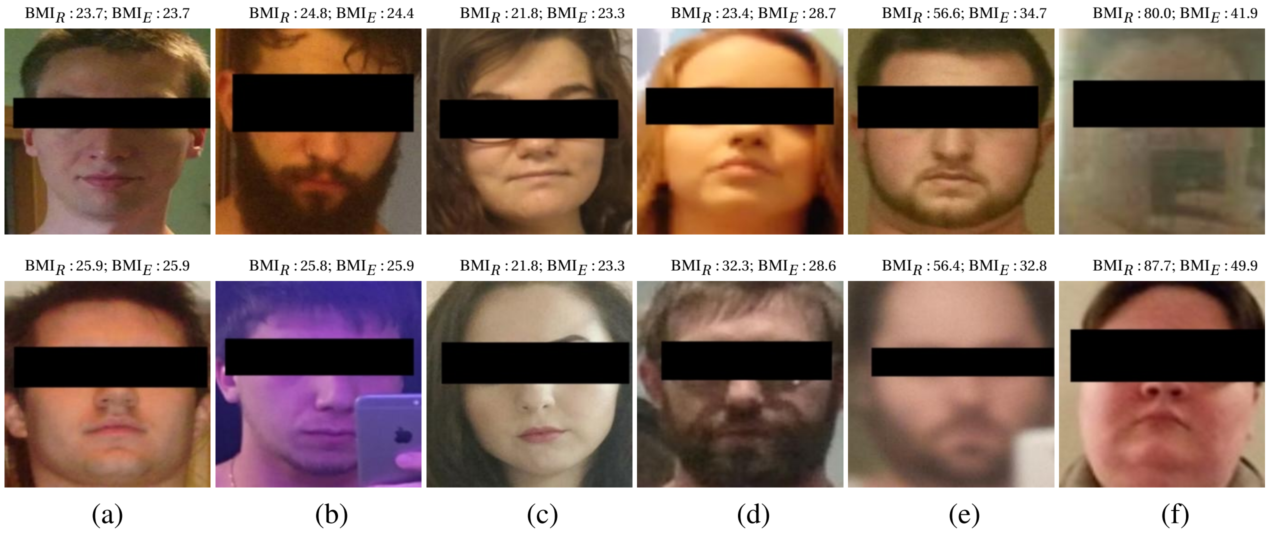


Figure 4: Samples of facial images from the VisualBMI dataset with both their ground truth (BMI_R) and estimated BMI value (BMI_E) respectively. Column (a) shows examples where BMI was perfectly estimated while the last column (g) shows examples with high error between the real and estimated BMI values.

4.1 Bayesian vs Non Bayesian ViT

Both ViT architectures (with and without Bayesian components) were trained using the same hyper-parameters which including batch size equals to 8, AdamW optimizer, *WeightedRandomSampler* functions to balance data according to gender and image resolution set to 224×224 pixels. An ensemble of 4 identical models were considered in each case (trained using different seeds).

Table 2 shows the BMI values estimated for each model and classified according to BMI categories (Normal, overweight, Obesity I and Obesity II) and Gender (Male and female). Results show that deterministic ViT outperforms the Bayesian variant slightly with MAE values of 4.375 and 4.434 for ViT and Bayesian-ViT respectively (only +0.0595 kg/m² of difference). The main contribution of the Bayesian approach is the uncertainty-based selective prediction.

In both cases BMI estimates for female subjects had worse performance than BMI for males. Such results are not surprising since the database had more male subjects than female. The test dataset in particular, contains 68% male and only 32% female subjects. Figure 4 shows several examples of facial images form VisualBMI dataset with their respective BMI ground truth value (BMI_R) and the estimated value (BMI_R) by the proposed Bayesian ViT architecture.

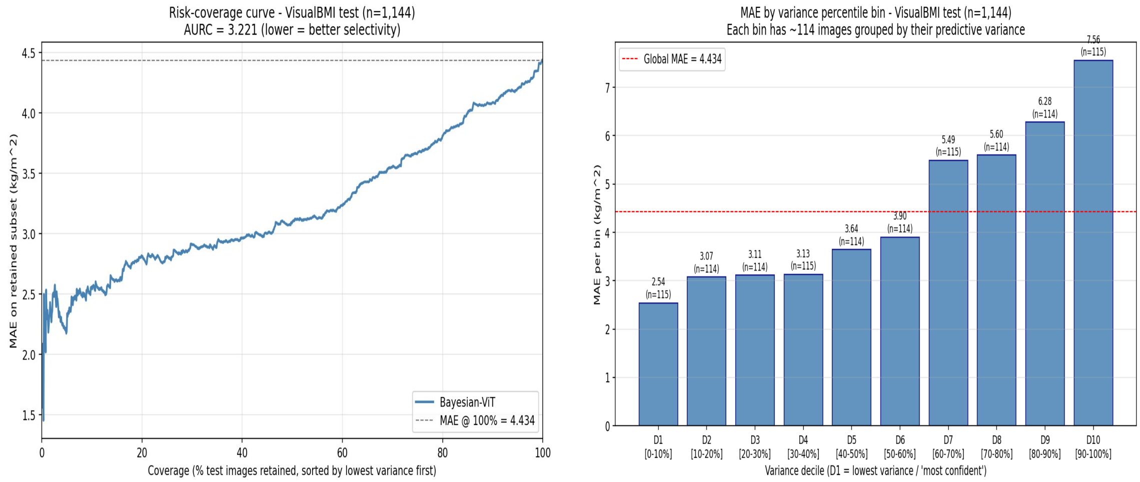
4.2 Uncertainty analysis

Results in Table 2 have shown that adding a Bayesian top to the ViT backbone does not improve the results of the deterministic ViT Model. However, the advantage of using the Bayesian model relies on the possibility of predicting estimate uncertainty. Therefore, the results can be classified according to how certain the model is of the BMI prediction. Two metrics were evaluated to measure uncertainty: variance and entropy (Equation 1).

Table 3 shows the MAE obtained when selecting BMI results according to variance. Predictions with lower variance have less uncertainty and are, therefore, more trustworthy. The test data were classified in percentiles (10,25,50,75 and 100) according to its predicted variance and computed MAE. The results show smaller MAE for the classes Normal and Overweight while Obese I and II present higher MAE values. This result confirms that these two classes are more difficult to predict since their distribution is not uniform. Therefore, the average error in these classes tends to be higher. An example of this tendency is shown in the MAE computed for the Obesity class II which correspond to 10.465 (the highest in the whole dataset). This results correspond to

Method	Normal	Overweight	Obesity I	Obesity II	Male	Female	σ^{2*}
Bayesian-ViT (10%)	1.927	2.577	7.435	10.462	2.579	2.483	2.560
Bayesian-ViT (25%)	2.125	2.149	5.699	10.382	2.743	3.079	2.814
Bayesian-ViT (50%)	2.718	1.874	4.430	9.785	2.941	3.546	3.100
Bayesian-ViT (75%)	3.112	2.129	3.430	9.040	3.420	4.257	3.668
Bayesian-ViT (100%)	3.361	2.497	3.400	8.196	4.073	5.209	4.434

Table 3: MAE obtained for each method computed by class and gender when using variance as discriminator for uncertainty. The overall MAE computed is displayed at the last column.



(a) Risk-Coverage Curve (b) MAE By Variance percentile bins

Figure 6: Figure shows the Risk coverage Curve in (a) where the MAE is computed as a function of the number of test images that are selected. Figure (b) shows the MAE as a function of the variance expressed in percentiles

only two samples where the BMI was poorly estimated. However, given the spread of the BMI values in this category, the MAE tends to be bigger and has a greater influence in the overall error. Figure 5 shows the specific images corresponding to this error in the Obesity Class with their respective BMI (real and estimated).

Figure 6 (a) shows the Risk Coverage Curve to evaluate how well an uncertainty measurement (in this case the variance) can identify unreliable predictions. The plot shows the increase in the MAE when more samples with less certainty are being added (x-axis). When all data is considered, the overall MAE is 4.434 as reported in Table 3.

$\text{BMI}_R : 36.97$; $\text{BMI}_E : 25.95$ $\text{BMI}_R : 36.91$; $\text{BMI}_E : 27.01$

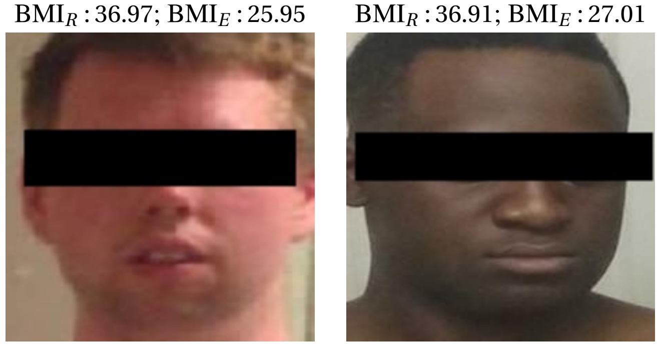


Figure 5: This two images correspond to the Obesity II class with a MAE of 10.462 reported in Table 3 .

The plot in Figure 6 (b) shows the MAE of the estimated BMI by variance percentile. As can be shown, the MAE increases when the variance increases showing the relation of the predicted variance with the error of the prediction and therefore, demonstrate the capability of this estimator as discriminator for better estimation results. Entropy has also been shown to be a good discriminator for uncertainty as is shown in column 9 of Table 3. In this case, the MAE computed is similar to the MAE achieved by the variance metric.

4.3 Comparison with the state of the art

The MAE of the estimated BMI for the test dataset is compared to the state of the art in Table 4. Only methods using the same database (visualBMI) are reported. Results show that the proposed method gives better results than most previous methods. There are three main differences between the proposed method and previous work: First, it is one of the first methods reporting the use of a transformer-based architecture where attention head is used instead of convolutional neural networks. Secondly, a Bayesian approach is added to quantify the uncertainty of the prediction and make the algorithm able to select only the most trustworthy results. Finally, the Bayesian ViT architecture was pre-trained using standard images (ImageNet-21k and ImageNet-1k). Therefore, no additional face datasets such as VGGFace, MS-Celeb or CASIA-WebFace were needed.

5 Conclusion and Future Work

A Bayesian-Vit Model to estimate the Body Mass Index (BMI) in the benchmark VisualBMI dataset has been presented. These results are comparable to the state of the art with the advantage that the certainty of the prediction can be quantified. The overall MAE in the ViT and Bayesian ViT was 4.375 and 4.434 respectively.

When selecting only the best predictions (with less uncertainty) the MAE is considerable reduced (up to 2.560 when considering the best 10% of the data and 3.668 when selecting 75%. BMI estimation using a single face image is still a challenging problem and there are several issues that still need to be addressed such as an imbalance in the datasets. For future work, the use of a classification model combined with regression to better estimate obesity classes will be explored. The use of ensemble Bayesian model and metrics for performing the ensemble will also be studied.

Paper	MAE
[Kocabey et al., 2017]	5.160
[Siddiqui et al., 2020]	5.020
[Aarotale et al., 2023]	5.130
[Yousaf et al., 2021]	4.990
Proposed ViT	**4.375**
Proposed Bayesian ViT	**4.434**
[Haritosh et al., 2019]	3.800
Proposed Bayesian ViT (75%)	**3.668**
Proposed Bayesian ViT (50%)	**3.100**
Proposed Bayesian ViT (25%)	**2.814**
Proposed Bayesian ViT (10%)	**2.560**

Table 4: Comparison of the proposed method with state of the art BMI estimation algorithms using single facial images and the VisualBMI dataset

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